Can AI-Human interaction drive sound change? Insights from phonetic accommodation in Hong Kong Cantonese [ŋ-]-[Ø-] onset merger

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Abstract

Traditionally, sound change is initiated, driven, and propagated exclusively through human-human interaction. However, with the increasing use of and access to AI-generated voices in recent years, their potential impact on ongoing sound change remains unclear. This study addresses this gap by investigating the [ŋ-]-[Ø-] onset merger in Hong Kong Cantonese. 33 native Hong Kong Cantonese speakers were recruited for a speech shadowing task where they were exposed to both customized AI-generated and human voices. Phonetic accommodation was assessed through temporal adjustment of the proportion of vowel within the target words. Applying the linear combination method, our findings revealed that phonetic convergence occurred in both shadowing and post-shadowing sessions for both voice sources. While the convergence effect was stronger for human voice during the shadowing session, the effects of the two voices did not differ significantly during the post-shadowing session. This implies that although human-human interaction may induce a stronger immediate convergence effect, the analysis failed to detect a difference between interaction conditions shortly after the shadowing task. This suggests that human-AI interaction has the potential to drive mini-sound change, accelerating or decelerating the ongoing process depending on the variants played.

Index Terms: sound change, human-AI interaction, phonetic accommodation, temporal adjustment

1. Introduction

In recent years, human speech interaction has become increasingly complex due to the rapid advancements in speech technology, especially with the development of Artificial Intelligence (AI)-generated speech sounds. Prior studies have largely focused on how human talkers modulate their speech patterns by comparing those in human-AI interactions with those in human-human interactions (e.g., [1], [2], [3]). Modulations have been found, including but not limited to changes in F0, duration, and vowel space. While these changes reflect how AI affects human's phonetic realizations during those turns of conversation, little is known about how these changes are retained to affect future speech behaviour, especially in the context of sound change.

Before the wide application of synthesized voice in the society, sound change is a process primarily driven by humans in the language community through intricate speech processing interactions among perception, production, and social factors (see [4] for more discussions). In contemporary language settings, however, humans are no longer just exposed to speech spoken by humans, as there is an increasing usage and chance of access to different artificial intelligence (AI) speech systems, ranging from commercialized to personalized ones (e.g., [5], [6], [7]). The advancement in technology allows synthesized speech sounds to be easily generated by machines to closely resemble human speech, and studies on speech sound detection suggested human may not possess the ability to correctly differentiate between AI and human voices ([8], [9]). These findings not only reopen questions about how humans as listeners process speech sounds, but also raise new questions about how processing affects subsequent speech production to drive potential sound change, particularly in the AI era. Adopting Ohala's view on the listener's role in sound change, when listeners fail to accurately reconstruct the non-canonical pronunciation, this failure, manifested as hypocorrection or hypercorrection, will lead to sound change when these listeners become speakers and produce their inaccurately reconstructed (non-canonical) forms [10]. Thus, it highlights a critical question: would speech behaviour differ if humans listen to and process AI-generated voices differently from human voices, eventually resulting in a distinct trajectory of sound change?

To experimentally examine sound change from this point of view, the speech shadowing paradigm, which assesses phonetic accommodation, has been employed as an effective approach in previous work (e.g., [4], [11], [12]). This is because both sound change and shadowing involve the underlying mechanisms of adjusting listeners' own production after exposure to a sound. [12] recently examined the phonetic accommodation of tone mergers in Hong Kong Cantonese (HKC) to AI and human talkers. Their findings showed that, when exposed to the new variants, these speakers converged their lexical tone to the new variants in both AI and human conditions. More importantly, a stronger degree of convergence was found when these speakers were in the human condition.

While their study revealed an effect of voice source (i.e., human vs. AI-generated voices) on ongoing sound change based on lexical tones, examining other acoustic features is essential to provide a more comprehensive picture of how different voice sources shape this process. The current study specifically investigates the [ŋ-]-[Ø] onset merger, the only

ongoing sound change in HKC that is traditionally conditioned by tone register. Its unique phonological characteristic makes it an ideal testing ground to examine the interaction between phonetics and phonology in the context of sound change. In HKC, [ŋ-]-[Ø-] onset merger has been a longstanding merger, of which its development can be traced back to at least 100 years ago (as cited in [13]). Traditionally, these two onsets were in a complementary distribution that they occur in different lexical tone registers, where /ŋ-/ occurs in a low tone register (low-falling^{21}, low-rising^{23}, and low-level$^{22/2}$ tones) while /Ø-/ in mid and high tone registers (mid-level$^{33/3}$, high-level$^{55/5}$, and high-rising^{25} tones) [14]. This distribution seems to have vanished over time. In [13]'s synchronic study of sound changes in Hong Kong with data collected around the mid-2000s, approximately 30% of adults (age 18-45) produced the opposite variant, regardless of the historical form (/ŋ-/ → [Ø-]: 37.5%; /Ø-/ → [ŋ-]: 31.3%). More recent work examining this onset merger with data collected around mid-2010s revealed that 76% of participants (age 17-60) used [ŋ-] regardless of the historical form, and no participant had a categorical contrast in the perception test [15]. Crucially, individual-level analysis showed that perception and production were not linked for this feature, unlike the two other sound change features in the same study with moderate positive correlation. The authors attributed this to the unique distribution (allophonic) and directionality (bidirectional) of this merger among other sound change features in HKC. Regardless of the underlying cause, this previous claim that the production-perception link does not exist may further complicate the assumed role of listeners as the initiators of sound change.

Although conflicting production results of these two studies may stem from differences in the data collection periods (mid-2000s vs. mid-2010s), our data can offer a synchronic, acoustic perspective on this diachronic issue. Collected in 2025 using six characters of this pair of sounds, our data is highly comparable (two in [13]; eight in [15]). By collecting baseline production data from participants in the shadowing paradigm, our study also contributes a synchronic update on this sound change feature in the mid-2020s.

By accounting for the effect of AI-generated voice on the phonetic accommodation of sound change feature [ŋ-]-[Ø-] onset merger using a speech shadowing paradigm, this study investigates how voice sources impact the spread of sound change in the AI era. Specifically, this study asks: 1. Does phonetic accommodation occur in [ŋ-]-[Ø-] onset merger? 2. Does the pattern differ between interacting with AI and human voices? 3. Does the pattern differ between immediate shadowing and post-shadowing?

2. Method

2.1. Participants

Thirty-three young native HKC speakers (21.52±3.76; 17F & 16M) were recruited from a university in Hong Kong. They were all born and raised in Hong Kong, and reported normal hearing and no history of speech disorders or cognitive disabilities. A consent form and a language background questionnaire were completed before the experiment.

2.2. Stimuli

Being a part of a larger project on HKC sound change accommodation, this study focuses on the [ŋ-]-[Ø-] onset merger. Six Chinese characters were selected to be the target stimuli of this feature, as listed in Table 1. Since the target stimuli were pre-recorded prior to the experiment, four out of six characters were produced with the [Ø-] onset to facilitate phonetic accommodation, given that recent work suggested that [ŋ-] onset has become more frequent in this ongoing sound change [15].

The human stimuli were recorded by two phonetically-trained, native HKC female research assistants (RA_1 & RA_2) in a sound-attenuated room using a Rode Lavalier GO microphone. The AI stimuli were generated using duration-informed TTS model [16] trained on 4000 Cantonese sentences recorded by each of the same two RAs. While this approach preserved the naturalness and voice characteristics to a large extent, listeners may still perceive the small differences introduced by the synthesis process. By training our own speech model, it allows us to control the variants produced in AI recordings, unlike previous studies using non-customized voices (e.g., [3], [17]) which they could not manipulate.

Table 1: *Target items examined in this study.*

Stimuli	Meaning	Shadowing Variant Played
我	'I'	[ŋɔ:^{23}]
餓	'hungry'	[ŋɔ:^{22}]
外	'outside'	[ɔ:i^{22}]
哀	'sorrow'	[ɔ:i^{55}]
愛	'love'	[ɔ:i^{33}]
曖	'ambiguous'	[ɔ:i^{25}]

2.3. Data Collection

Following the shadowing paradigm [18], participants completed pre-shadowing (baseline), shadowing, and post-shadowing sessions. Throughout the entire experiment, the participants were instructed to speak naturally. In both the pre-shadowing and the post-shadowing sessions, participants were presented with a wordlist consisting of the target characters along with other non-target filler words. During the shadowing session, the target character was displayed on the computer screen along with an image (either an avatar face or a human face) and an audio recording of that character. The participants were instructed to read the characters after hearing the model talker's recordings and were given no information regarding whether to follow the pronunciation or not from the recordings.

This study used a between-subject factor for *Face* and within-subject factors for *Voice* and *the identity of the model talker* (RA_1 or RA_2). Participants were randomly assigned to one of the conditions. In other words, approximately half of the participants interacted with an AI face during the shadowing session, while the other half interacted with a human face. Within each group, the presentation order of voice (either AI voice first or human voice first) was counterbalanced. Each participant was expected to produce 30 target tokens of the [ŋ-]-[Ø-] merger (6 target words in baseline×1 + shadowing×2 + post-shadowing×2), resulting in a dataset of 990 tokens.

After completing the three sessions, participants filled out a post-questionnaire regarding their thoughts about the experiment, their awareness of the manipulation, and their experience on the use of AI technology.

2.4. Data Processing

Recordings of both the stimuli and participants responses were screened to exclude incomplete or mispronounced tokens (11.01%) in Praat [19]. Vowel proportion was selected as the dependent variable as this acoustic correlate provides temporal information about onset realization, reflecting the degree to which an overt onset is reduced to a null onset variant, or conversely, the extent to which a null onset is realized as an overt onset variant in CV-structured monosyllabic words. Each character was segmented, and the absolute duration of the segments was extracted. Then, the vowel proportions were calculated and standardized (centered & scaled) in one group to preserve the relational structure of the data [20]. Assuming phonetic convergence, vowel proportions would shorten when exposed to shorter vowel proportions ([ŋ-]-like variant) uttered by the model talker, and lengthen when exposed to longer ones ([ø-]-like variant).

To assess whether phonetic accommodation occurred, this study employed the linear combination method [21] rather than the traditional difference-in-difference (DID) approach, which has been shown to be biased [20]. It uses regression to calculate accommodation. With baseline values being added to evaluate participants' consistency across sessions, the model talkers' values, along with the factors manipulated, are included to evaluate potential convergence effects. This prevents biases where extreme baseline values and small initial differences overestimate convergence and divergence respectively. Specifically, two models were fitted in R using the lme4 and lmerTest packages ([22], [23], [24]):

*Shadowing (3.1.1.) / Post-Shadowing (3.1.2.) ~ Baseline + ModelTalker*Voice*Face + ModelTalkerID($RA_{1/2}$) + (1|ParticpantID) + (1|Character) + (0 + ModelTalker| ParticipantID)*

3. Results

3.1. Phonetic Accommodation

ANOVA Type III test with Satterthwaite's approximation was used to analyze the results of each linear mixed-effect model. The alpha level for significance was set at 0.05. To ensure model robustness, multi-collinearity, singularity, homoscedasticity, and normality were visually and statistically examined using the performance package [25]. Results of the models were visualized using the ggplot2 package [26].

3.1.1. Shadowing

The model revealed a significant effect of Baseline ($F(1, 247.92) = 22.64$, $p < .001$), ModelTalker ($F(1, 13.85) = 72.51$, $p < .001$), and Voice ($F(1, 276.37) = 8.23$, $p = .004$)). These suggested not only that participants' performance remained consistent across tasks, but also their speech tended to converge acoustically towards the variants played, supported by the significant positive effects of Baseline and ModelTalker respectively (shown in Figures 1 & 2).

Moreover, the effect of ModelTalker changed depending on Voice, as shown in the significant two-way interaction ($F(1, 276.51) = 12.64$, $p < .001$). To further examine the interaction, a post-hoc slope analysis was conducted. While significant relationships were found for both voices (AI voices: $\beta = .38$, $SE = .07$, 95% CI [0.24, 0.51], $t(27.1) = 5.74$, $p < .0001$; Human voices: $\beta = .62$, $SE = .07$, *95% CI* [0.48, 0.77], $t(26.4) = 8.75$, $p < .0001$), the relationship was stronger on the human voices ($\beta = -.25$, $SE = .07$, *95% CI* [-0.38, -.11], $t(286) = -3.55$, $p < .001$). These results demonstrated that they were more strongly influenced by human speech during shadowing, as shown in Figure 2. No other main effects or interactions were significant.

3.1.2. Post-Shadowing

The model revealed a significant effect of both Baseline ($F(1, 273.92) = 20.34$, $p < .001$) and ModelTalker ($F(1, 5.96) = 13.31$, $p = .01$). However, the two-way interaction is not significant ($F(1, 290.54) = 0.03$, $p = .87$)) between ModelTalker and Voice. These demonstrated that after the stimuli exposure period, participants' performance remained consistent across tasks (as shown in Figure 3). Moreover, their tendency to converge to the target variants persisted, even though the analysis failed to detect a significant effect of voice on convergence (as shown in Figure 4). No other main effects or interactions were significant.

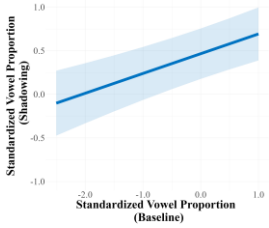


Figure 1: *Effect of participants' baseline vowel proportions on their shadowed values.*

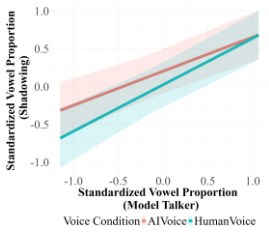


Figure 2: *Effect of model talkers' vowel proportion on their shadowed values.*

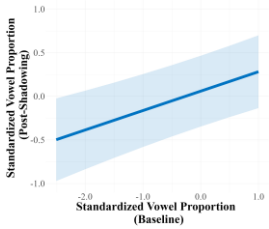


Figure 3: *Effect of participants' baseline vowel proportions on their post-shadowed values.*

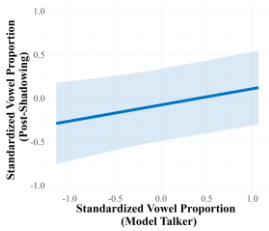


Figure 4: *Effect of model talkers' vowel proportion on their post-shadowed values.*

3.2. A Synchronic Analysis of [ŋ-]-[Ø-]

The baseline vowel proportion of characters across their respective tone register was examined. Rather than directly comparing our acoustic results with the previous perceptual findings ([13], [15]), this section aims to expand our understanding of this merger in the mid-2020s.

A two-sample Kolmogorov Smirnov test was performed using the KSgeneral package [27] to examine if the usage proportion of [ŋ-] or [Ø-] differed based on the tone register. The results showed that the distributions significantly differ from each other ($d = 0.24$, $p < .0001$), indicating that the usage proportion of [ŋ-] or [Ø-] remains bounded by tone register acoustically.

In other words, compared to characters in high and mid tone registers, those in low register tend to exhibit a shorter vowel proportion and are more likely to be produced with the [ŋ]-like variant, as shown in Figure 5 (see where the red arrow points to).

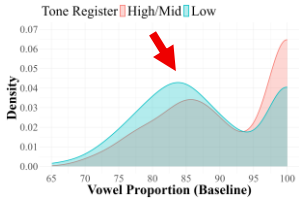


Figure 5: *Distribution of vowel proportion.*

4. Discussion

This study investigated whether phonetic accommodation of the ongoing [ŋ-]-[Ø-] onset merger in HKC differs between human-human and human-AI interactions. Phonetic convergence occurred in both shadowing and post-shadowing sessions, evidenced by the significant positive effects of the model talkers. However, the effects of AI voice and human voices differed between these two sessions. While participants demonstrated greater sensitivity to the human voice than the AI voice in the shadowing session, no difference was detected in the post-shadowing session.

These results suggest that participants exhibited different speech behaviours depending on the voice during immediate interaction, with them converging more strongly towards other human speakers. However, the leftover effect brought by this convergence showed no significant difference over time. While these results partially align with previous work on lexical tone that convergence effect is stronger towards humans [12], this study expands current knowledge by revealing that, regardless of the voice source, phonetic convergence introduced during shadowing can be retained to some extent shortly after the exposure. This suggests comparable effect of AI and human voices over time in affecting speech behaviour.

Aligning with Ohala's view on the listener's role in sound change [10], our results showed that a 'mini sound change' occurred both during and after shadowing. The short-term carryover accommodation effect may reflect a long-term sound change process, although we acknowledge the post-shadowing data were conducted shortly after the shadowing task. If the voice source only affects the immediate degree of convergence (more sensitive to human voice), it implies that both AI and human voices can potentially drive sound change in progress in the long run.

Given our increasing exposure to AI-generated voices, this further indicates that synthesized speech may accelerate or decelerate the progress of ongoing sound changes, depending on which variants it is more likely to generate. As existing AI voices mostly produce the 'standard' forms of the sound change features (i.e., [Ø-] in both high and mid tone registers and [ŋ-] in low tone register in this study), the emerging form could be suppressed or even reversed.

Moreover, while recent work showed that no relationship between perception and production was found for this sound change feature from the perceptual findings [15], our study showed that participants could successfully converge towards the model talker through the temporal adjustment of vowel proportion. This implies that some underlying linkage may still exist. Also, our study did not find a consistently higher usage proportion of [ŋ-] across characters in different tone registers. Contrary to the previous finding [15], characters in high or mid tone register are more likely to be produced with the [Ø-]-like variant than those in low tone register. Although all characters were produced with both variants at the group level, with variation across individual speakers, the distribution remains bounded by tone register, as evidenced by the two distinct distributional patterns in mid-2020s.

While this study considered whether participants were aware of the AI voices, the sample size within each subcategory was unbalanced and too small for statistical modelling. Among the 33 participants, 13 correctly differentiated the human voices and AI voices, with a higher accuracy rate observed among participants who interacted with the human face, as shown in Table 2. Crucially, these findings offered an insight that visual cues (Face) can potentially influence how participants identify a voice source (Voice), subsequently affecting voice processing and shadowing behaviour. This possibility warrants further investigations to examine if explicit awareness of voice sources modulates the patterns of phonetic accommodation.

Table 2: *Distribution of identification of voice.*

		Human Face (n = 16)	AI Face (n = 17)
Correct Identification		8 (50.0%)	5 (29.4%)
Misidentification	Both AI	3 (18.8%)	4 (23.5%)
	Both Human	5 (31.3%)	8 (47.1%)

Lastly, this study has several additional limitations, including considerable data attrition during data cleaning, reliance on a single acoustic correlate, and a limited sample size for testing three-way interactions. As part of a larger research project, future work is underway to expand the sample size, examine additional acoustic correlates associated with different sound change features, and include speakers from a broader range of generations to further examine intergenerational and cross-feature effects on accommodation patterns in shaping the sound change trajectory in the AI era.

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6. References

[1] L. Bradshaw, V. Perepelytsia, and V. Dellwo, "Vocal effort in human interactions with voice-AI," in *Proceedings of ICPhS*, 2023. Accessed: May 20, 2026. [Online]. Available: https://www.zora.uzh.ch/server/api/core/bitstreams/fc399582-3122-471a-86b6-9a6432e0be9d/content

[2] X. Chen *et al.*, "Modulation of Phonetic Realizations in Cantonese Dialogue with Human and AI Interlocutors," presented at the Proc. Interspeech 2026, Sydney, Australia, Sep. 2026.

[3] M. Cohn, B. F. Segedin, and G. Zellou, "Acoustic-phonetic properties of Siri- and human-directed speech," *J. Phon.*, vol. 90, p. 101123, Jan. 2022, doi: 10.1016/j.wocn.2021.101123.

[4] A. Garrett and K. Johnson, "Phonetic bias in sound change," in *Origins of Sound Change: Approaches to Phonologization*, A. C. L. Yu, Ed., Oxford University Press, 2013, p. 0. doi: 10.1093/acprof:oso/9780199573745.003.0003.

[5] S. Creer, S. Cunningham, P. Green, and J. Yamagishi, "Building personalised synthetic voices for individuals with severe speech impairment," *Comput. Speech Lang.*, vol. 27, no. 6, pp. 1178–1193, Sep. 2013, doi: 10.1016/j.csl.2012.10.001.

[6] L. Dai, V. Kritskaia, E. van der Velden, M. M. Jung, M. Postma, and M. M. Louwerse, "Evaluating the usage of Text-To-Speech in K12 education," in *Proceedings of the 2022 6th International Conference on Education and E-Learning*, in ICEEL '22. New York, NY, USA: Association for Computing Machinery, 2023, pp. 182–188. doi: 10.1145/3578837.3578864.

[7] L. Jia *et al.*, "The Impact of AI Speech Synthesis on the Broadcasting Profession and Its Transformation Path: A Study Based on TTS Technologies," in *Proceedings of the 2025 International Symposium on Artificial Intelligence and Computational Social Sciences*, in ACM Other conferences. , 2025, pp. 277–281. doi: 10.1145/3776759.3776803.

[8] S. Barrington, E. A. Cooper, and H. Farid, "People are poorly equipped to detect AI-powered voice clones," *Sci. Rep.*, vol. 15, no. 1, p. 11004, Mar. 2025, doi: 10.1038/s41598-025-94170-3.

[9] K. T. Mai, S. Bray, T. Davies, and L. D. Griffin, "Warning: Humans cannot reliably detect speech deepfakes," *PLOS ONE*, vol. 18, no. 8, p. e0285333, Aug. 2023, doi: 10.1371/journal.pone.0285333.

[10] J. J. Ohala, "The listener as a source of sound change: An update," in *The Initiation of Sound Change*, John Benjamins Publishing Company, 2012, pp. 21–36. Accessed: Jun. 07, 2026. [Online]. Available: https://www.degruyterbrill.com/document/doi/10.1075/cilt.323.05oha/html?srsltid=AfmBOoq-oCuYF1BNQZyEMyBPX2g-oSWASSegK_nTMoLHX4I7D53C15bQ&lang=en

[11] Y. Lin, Y. Yao, and J. Luo, "Phonetic accommodation of tone: Reversing a tone merger-in-progress via imitation," *J. Phon.*, vol. 87, p. 101060, Jul. 2021, doi: 10.1016/j.wocn.2021.101060.

[12] Y. Hong, Y. Tian, G. W. Cao, T. Lee, and P. P. K. Mok, "Phonetic Accommodation to AI and Human talkers: A Case Study of Cantonese Tone Mergers," presented at the Proc. SpeechProsody 2026, 2026, pp. 159–163. doi: 10.21437/SpeechProsody.2026-32.

[13] C. K. S. To, S. Mcleod, and P. S. P. Cheung, "Phonetic variations and sound changes in Hong Kong Cantonese: diachronic review, synchronic study and implications for speech sound assessment," *Clin. Linguist. Phon.*, vol. 29, no. 5, pp. 333–353, May 2015, doi: 10.3109/02699206.2014.1003329.

[14] R. S. Bauer and P. K. Benedict, *Modern Cantonese phonology*. De Gruyter Mouton, 1997. doi: 10.1515/9783110823707.

[15] L. S. P. Cheng, M. Babel, and Y. Yao, "Production and perception across three Hong Kong Cantonese consonant mergers: Community- and individual-level perspectives," *Lab. Phonol.*, vol. 13, no. 1, Jun. 2022, doi: 10.16995/labphon.6461.

[16] C. Yu *et al.*, "DurIAN: Duration Informed Attention Network for Speech Synthesis," presented at the Proc. Interspeech 2020, 2020, pp. 2027–2031. doi: 10.21437/Interspeech.2020-2968.

[17] M. Cohn and G. Zellou, "Prosodic Differences in Human- and Alexa-Directed Speech, but Similar Local Intelligibility Adjustments," *Front. Commun.*, vol. 6, Jul. 2021, doi: 10.3389/fcomm.2021.675704.

[18] S. D. Goldinger, "Echoes of echoes? An episodic theory of lexical access," *Psychol. Rev.*, vol. 105, no. 2, pp. 251–279, 1998, doi: 10.1037/0033-295X.105.2.251.

[19] P. Boersma, D. Weenink, and A. Shchupak, *Praat: doing phonetics by computer [Computer program]*. (2026). [Online]. Available: https://praat.org

[20] B. MacLeod, "Problems in the Difference-in-Distance measure of phonetic imitation," *J. Phon.*, vol. 87, p. 101058, Jul. 2021, doi: 10.1016/j.wocn.2021.101058.

[21] U. Cohen Priva and C. Sanker, "Limitations of difference-in-difference for measuring convergence," *Lab. Phonol.*, vol. 10, no. 1, Sep. 2019, doi: 10.5334/labphon.200.

[22] A. Kuznetsova, P. B. Brockhoff, R. H. B. Christensen, and S. P. Jensen, *lmerTest: Tests in Linear Mixed Effects Models*. (Mar. 05, 2026). Accessed: Jun. 09, 2026. [Online]. Available: https://cran.r-project.org/web/packages/lmerTest/index.html

[23] D. Bates *et al.*, *lme4: Linear Mixed-Effects Models using "Eigen" and S4*. (Mar. 05, 2026). Accessed: Jun. 09, 2026. [Online]. Available: https://cran.r-project.org/web/packages/lme4/index.html

[24] Posit team, *RStudio: Integrated Development Environment for R*. Boston, MA: Posit Software, PBC, 2025. [Online]. Available: http://www.posit.co/

[25] D. Lüdecke *et al.*, *performance: Assessment of Regression Models Performance*. (May 21, 2026). Accessed: Jun. 09, 2026. [Online]. Available: https://cran.r-project.org/web/packages/performance/index.html

[26] H. Wickham *et al.*, *ggplot2: Create Elegant Data Visualisations Using the Grammar of Graphics*. (Apr. 22, 2026). Accessed: Jun. 09, 2026. [Online]. Available: https://cran.r-project.org/web/packages/ggplot2/index.html

[27] D. S. Dimitrova, Y. Jia, V. K. Kaishev, and S. Tan, *KSgeneral: Computing P-Values of the One-Sample K-S Test and the Two-Sample K-S and Kuiper Tests for (Dis)Continuous Null Distribution*. (Jul. 29, 2024). Accessed: Jun. 09, 2026. [Online]. Available: https://cran.r-project.org/web/packages/KSgeneral/index.html